

ASSIGNMENT SET – 1

Q1.

Trends in Operations Management

Operations Management has seen tremendous changes recently. The following are some of the emerging trends influencing organizations' operations:

1. Automation and Industry 4.0

- **Smart factories:** Robots, artificial intelligence, and IoT sensors used to automate repetitive activities
- **Real-time information sharing:** Communication between machines and gathering of continuous data
- **Predictive maintenance:** Maintenance of equipment that predicts problems before they occur
- **Lowering of costs:** Automation removes inefficiencies and decreases Labor expenses

2. Sustainability and Sustainable Operations

Restructuring of the supply chain: Minimization of environmental footprint throughout the value chain

- **Sustainable energy:** Using green sources like wind and solar to manufacture products
- **Waste reduction:** Eliminating any wastage during the process of production
- **Competitive edge:** Sustainability becomes a competitive advantage due to customers' preference

3. Globalization and Complexities in Supply Chains

Sourcing from several countries: Parts from different countries are acquired to gain cost advantage

- **Introduction of new risks:** Challenges due to long lead times, political instabilities, and cultural differences
- **Disruptions in the supply chain:** The pandemic exposed weaknesses of globalization
- **Disruption management:** Diversification of supplier base

4. Lean Manufacturing

- **Eliminate waste:** Remove excess inventory, defects, and unnecessary motions
- **Cost savings:** Operational costs greatly reduced through lean principles
- **Quality improvement:** Used with Six Sigma for improved quality control
- **Continuous improvement:** Process improvements ongoing

5. Mass Customization

- **Balancing efficiency and customization:** Produce customized goods on large-scale
- **Dell computer case:** Customer chooses configuration and company produces
- **Extended use:** Applies now to footwear, clothes and even medications

- **Improved customer satisfaction:** Due to customization

6. Digital Revolution & IoT

- **Connected sensors:** Information received in real time from facilities, vehicles and machinery
- **Data-driven decision-making:** Decisions are made by operations manager in real-time with data
- **Optimization of routes:** Route optimization is carried out with help of data
- **Forecasting:** Forecasting enhanced due to availability of continuous data

7. Nearshoring and Reshoring

- **Cost increases overseas:** Labor cost increases in offshore countries
- **Speed benefit:** Closer proximity to the market facilitates faster production
- **Supply chain resilience:** Reduced supply chain disruptions due to closeness
- **Back reshoring:** Trend among companies to return production to country of origin

8. Employee Skill Enhancement

- **Skill demands:** Employees will require skills such as data analysis and management systems.
- **Investment in training:** A large investment is necessary from companies in training employees.
- **Managing change:** Organizational culture changes should be managed well during implementation.
- **Security of employment:** Employee concerns regarding job safety due to automation must be addressed.

Conclusion: All eight trends discussed above are interrelated and support each other. Automation needs high-quality processes to succeed. Digital transformation leads to lean management. Nearshoring arises from the need for sustainability. Organizations that adopt all these trends will gain the edge over their competitors. The need is to integrate automation, sustainability, lean concepts, digital analytics, and supply chain management into organizational practices.

Q2.

Understanding of Operations Strategy

Operations strategy links business strategy to production and logistics. This is how firms implement their competitive position via operations.

Elements of Operations Strategy

- **Low-cost leadership:** Compete on being the lowest cost manufacturer
- **Quality orientation:** Compete based on providing high-quality goods
- **Responsiveness/speed:** Compete by being fast at everything
- **Flexibility:** Compete on ability to adapt rapidly to customer demands
- **Innovation:** Compete based on product improvements

Case Study of Competitive Operations Strategy

Strategic Operations Approach of Dell Computer:

- **Business strategy:** Provide cheap, readily available computers for everyone
- **Operations strategy:** Sell to consumers directly; customize and manufacture only upon order
- **Inventory policy:** Very low inventory levels, since customers order and Dell manufactures after that
- **Supply chain:** Extremely well-coordinated, especially regarding components' suppliers
- **Cost structure:** Tremendously cheaper operation compared to competitors who have to hold inventory
- **Consequence:** Dell's leadership position as a result

Strategic Approach Adopted by Apple Computer:

- **Business strategy:** Provide top-quality, innovative computers to wealthy consumers
- **Operations strategy:** Outsource production while designing and branding products
- **Focus area:** Stringent quality control on outsourced products
- **Brand experience:** Stores provide experience-oriented shopping, not efficient shopping
- **Positioning:** Consumers pay premium prices for the products offered
- **Consequence:** Apple became premium brand due to operations supporting quality and innovation

Importance of Operations Strategy

- **Implementation:** Execution of business strategy happens at the operations level
- **Cost implications:** Operations strategy determines profitability of a firm
- **Market positioning:** Operations helps in occupying market positions
- **Competitive advantage:** Enables/disables competitive advantage through operations
- **Criticality of alignment:** Non-alignment with operations results in failure

Conclusion: An operations strategy cannot be separate from a business strategy since it acts as its implementation mechanism. Dell thrived because of operations strategy that supported its positioning strategy by being cost-effective. On the other hand, Apple Computer thrives because of its operations strategy that focuses on quality and supports premium positioning strategy. Both companies have consciously aligned operations strategies with competitive advantage strategy.

Q3.

Qualitative Forecasting Methods

- **Delphi method:** Experts predict, discuss anonymously, and form consensus
- **Market research:** Ask consumers regarding their purchase intentions in the future
- **Judgmental forecasts:** Managers make estimates using experience and intuition
- **Suitable for:** Introduction of new products and changes in the market
- **Strengths:** Incorporates expert knowledge
- **Weaknesses:** Subjective, prone to politics, and less accurate than quantitative forecasting

Quantitative Forecasting Methods

- **Moving Average:** Calculation of the average demand of the previous N periods
- **ARIMA:** Statistical technique that analyses data trends and inter-relationships
- **Exponential smoothing: Recent observations are weighted more than distant observations**
- **Regression analysis:** Establish relationships between variables and demand

When to Use ARIMA Instead of Moving Average?

- **Presence of seasonality:** ARIMA accounts for seasonality, but moving average doesn't
- **Existence of trend:** For instance, when there is continuous increase in demand
- **Volatility:** If data have a volatility component, use ARIMA
- **Existence of auto-correlation:** When there is a connection between past and future values
- **Fashion retail:** Season of winter coats is November-December. Moving average will miss the peaks and troughs.

Effects on Capacity and Design

- **Accuracy is important:** Incorrect predictions result in incorrect capacity size
- **Risk of Moving Average:** Moving Average on seasonal information results in overestimation of capacity for low seasons and underestimation for high seasons
- **Benefit of ARIMA:** ARIMA shows that there are 8,000 units during low seasons and 14,000 during high seasons compared to the average 10,000
- **Capacity design approach:** Rather than using fixed capacity, use flexible one. Outsource during peak seasons. Utilize temporary employees
- **Consequence:** Correct predictions can help in making correct decisions regarding capacity which results in saving money spent on the wrong facilities
- **Consequences:** Decision-making regarding capacity will affect the bottom line for many years. Accuracy is therefore very important

Conclusion: Qualitative forecasting method should be used when dealing with strategic decisions and new products. Quantitative method can be used for operations management based on historical data. ARIMA should be preferred over moving average if demand has trend or seasonality. Given the large capital expenses for capacity and the consequences of those decisions for a long time, the proper forecasting method is very important to the bottom line.

ASSIGNMENT SET – 2

Q4.

Quantitative Location Considerations

- **Transportation cost:** Cost of transporting raw materials in and finished goods out
- **Labor cost:** Wage levels prevailing in different locations
- **Real Estate Cost:** Price of land and building varies considerably in different regions
- **Utility cost:** Electric power, water, etc. varies considerably in different places
- **Tax considerations:** Subsidies and tax incentives available in different regions
- **Break even analysis:** Compute cost and determine lowest cost locations

Non-quantitative Considerations (Usually more important than quantitative)

- **Branding:** “Made in USA” commands premium price for luxury items
- **Talent:** Silicon valley attracts the best talent justifying high wages
- **Product Quality:** Precision manufacturing requires costly talent in developed nations
- **Supply network:** Some regions have specialized suppliers’ cluster already formed
- **Political Stability:** Stable political and legal system makes enormous difference
- **Infrastructure:** Stable electricity supply and reliable internet/transportation infrastructure is crucial
- **Proximity to Customer:** Proximity allows quick response and relationship building
- **Innovative Capability:** Places such as silicon valley offer innovation not possible anywhere else

Reasons for Choosing Higher-Cost Sites

- **Coach company:** Produces in the expensive U.S.A., since “Made in USA” allows for high prices
- **Nike company:** Produces in Vietnam, where production costs are low; illustrates cost dominance in certain firms
- **Insurance:** Keep expensive sites just in case of disruption in primary locations
- **Fast fashion:** Proximity to markets is needed for speedy response
- **Quality:** Certain products need expensive labor for high-quality output
- **Consistency:** Existing supplier networks justify high costs

Factor Rating Technique

- **List all factors:** List all factors influencing location decision-making
- **Weight factors:** Determine weights for each factor in terms of strategy
- **Rate locations:** Rate each site on each factor
- **Sum up:** Weighted scores dictate optimal location choice
- **Strategy:** Encourages thinking about the most important aspects of strategy

Conclusion: Site selection decisions have to take into account both the quantifiable and strategic elements. Cost minimization alone will often result in strategically inferior site selections. Companies which consider branding, quality, innovation, consistency, and relations get better results than others. Companies which do not consider strategic factors will soon learn that cost advantages disappear once hidden costs arise.

Q5.

Just-in-Time Manufacturing

- **Basic principle:** Manufacture exactly what is required exactly when it is required
- **Limited stocks:** Stock only a few days of materials
- **Financial benefits:** No need for costly inventory storage
- **Less working capital:** Provides working capital for something else
- **Increased efficiency:** Wastage is reduced and efficiency is increased

Total Quality Management

- **Quality embedded:** Quality built into the process
- **Everyone responsible:** Everyone is responsible for quality
- **Continuous improvement:** Continuous process improvement
- **Prevention orientation:** Prevention instead of correction of defects
- **Reduced costs:** Fewer defects lead to cost reductions

The Critical Synergy

- **Inventory buffer:** Extra inventory takes care of 5% defect rates
- **JIT system weakness:** 5% defect rate will stop production
- **No inventory buffer:** No way to compensate for quality with extra inventory
- **Cascade effect:** One defect stops the whole chain
- **Requirement:** Perfect quality is mandatory for JIT

Process Capability

- **Measuring process capability:** Cp and Cpk measure process capability
- **Process capability requirements:** High process capability is needed in JIT
- **Risk of unpredictability:** Lack of process capability means lack of quality predictability
- **Impossible planning:** If quality is unpredictable, then it is impossible to plan

Lean Six Sigma: The Connecting Factor

- **"Lean" aspect:** Removal of wasteful non value adding inventory
- **"SIX SIGMA" aspect:** Variability is decreased through statistics
- **Combined effect:** System efficiency AND stability at once
- **The Toyota approach:** The first company to use them together
- **The current approach:** The top organizations use Lean Six Sigma together

Conclusion: JIT and TQM are indeed complementary and interlinked in theory. JIT needs good quality to operate safely; improvements in quality but without lean is just too costly. Lean Six Sigma offers a holistic approach that makes both effective together. Thus, one gets benefits of lower costs due to lower inventories and quality because of process improvement at the same time.

Q6.

Level Strategy: Continuous Production

- **Principle:** Produce and staff continuously throughout the year, independent of demand
- **Employment stability:** No layoffs or massive recruitment; provides job security for workers
- **Utilization of machinery:** Plants operating continually ensures return on capital investment
- **Fixed cost structure:** Consistent payroll and other expenses that are predictable
- **Improve quality:** Teams become proficient with stable labor force thus improve quality
- **Disadvantages:** High levels of inventory build up when there is low demand for excess working capital

Chase Strategy: Matching Demand

- **Principle:** Modify production and workforce according to demand fluctuations
- **Low inventories:** Produces only what is currently required
- **Low holding costs:** Low cost of warehouses and storage
- **Reacts to sudden demands:** Able to accommodate increased demand
- **Disadvantages:** High cost of hiring and training employees, layoff costs, quality problems

Operations in Manufacturing

- **Common choice:** Level strategy more often chosen
- **Machinery costs:** Expensive machinery used in manufacturing should be operated all the time
- **Set-up costs:** Expensive to stop and start production process
- **Product storage:** Products made in manufacturing can be stored
- **Inventory strategy:** Produce when demand is low and sell when demand is high

Service Organizations

- **Usual option:** Chase Strategy
- **Services are perishable:** any unused capacity is wasted – cannot be stored
- **Wide fluctuations in demands:** peak hours for restaurants are lunch and dinner time; for airlines - holiday periods
- **Labor is the key cost driver:** ability to be flexible with labor is essential
- **Examples:** airlines employ seasonal labor to manage peaks; hospitals change their staffing weekly

Hybrid Approaches

- **Manufacturer's strategy:** outsource during peak periods without hiring additional personnel permanently

- **Service strategy:** employ part-time labor for peaks to avoid costly layoffs
- **Flexibility improvement:** both approaches give necessary flexibility
- **Cost savings:** find a balance between employment stability and cost responsiveness

Conclusion: Level Strategy focuses on employment stability and efficiency; Chase Strategy focuses on cost responsiveness. Manufacturers prefer Level Strategy due to capital intensity of production; service organizations need Chase Strategy as they cannot inventory services. Both have trade-offs. Best firms strategically select their strategy taking into account their organization's features and competitive needs.